



GSFA Online

Updated up to the 48th Session of the Codex Alimentarius Commission (2025)

FOOD ADDITIVE GROUP DETAILS

PHOSPHATES

The provisions that follow are defined at the additive group level, and thus apply to the *total* content of the additives participating in this group. Additives that make up this group are provided for reference only.

Participating Additive(s)

INS No.	Additive Name
 342(i)	Ammonium dihydrogen phosphate
 452(v)	Ammonium polyphosphate
 542	Bone phosphate
 450(vii)	Calcium dihydrogen diphosphate
 341(i)	Calcium dihydrogen phosphate
 341(ii)	Calcium hydrogen phosphate
 452(iv)	Calcium polyphosphate
 342(ii)	Diammonium hydrogen phosphate
 450(vi)	Dicalcium diphosphate
 340(ii)	Dipotassium hydrogen phosphate
 450(i)	Disodium diphosphate
 339(ii)	Disodium hydrogen phosphate
 450(ix)	Magnesium dihydrogen diphosphate
 343(i)	Magnesium dihydrogen phosphate
 343(ii)	Magnesium hydrogen phosphate
 451(ii)	Pentapotassium triphosphate
 451(i)	Pentasodium triphosphate
 338	Phosphoric acid
 340(i)	Potassium dihydrogen phosphate
 452(ii)	Potassium polyphosphate
 452(iii)	Sodium calcium polyphosphate
 339(i)	Sodium dihydrogen phosphate
 452(i)	Sodium polyphosphate
 450(v)	Tetrapotassium diphosphate
 450(iii)	Tetrasodium diphosphate
 341(iii)	Tricalcium phosphate
 343(iii)	Trimagnesium phosphate
 340(iii)	Tripotassium phosphate
 450(ii)	Trisodium diphosphate
 339(iii)	Trisodium phosphate

CSV (Tables)

GSFA Provisions for PHOSPHATES

Number	Food Category	Max Level	Notes
 06.6	Batters (e.g. for breading or batters for fish or poultry)	5,600 mg/kg	 Note 33
 01.3.2	Beverage whiteners	13,000 mg/kg	 Note 33  Note 480
 07.1.4	Bread-type products, including bread stuffing and bread crumbs	9,300 mg/kg	 Note 229  Note 33
 06.3	Breakfast cereals, including rolled oats	2,200 mg/kg	 Note 33
 02.2.1	Butter	880 mg/kg	 Note 33  Note 34
 04.1.2.7	Candied fruit	10 mg/kg	 Note 33

Number	Food Category	Max Level	Notes
04.2.2.4	Canned or bottled (pasteurized) or retort pouch vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), and seaweeds	2,200 mg/kg	Note XS258R Note XS257R Note 33 Note XS57
06.5	Cereal and starch based desserts (e.g. rice pudding, tapioca pudding)	7,000 mg/kg	Note 33
01.6.5	Cheese analogues	9,000 mg/kg	Note 33
05.3	Chewing gum	44,000 mg/kg	Note 33
14.2.2	Cider and perry	880 mg/kg	Note 33
05.1.4	Cocoa and chocolate products	1,100 mg/kg	Note 33
05.1.1	Cocoa mixes (powders) and cocoa mass/cake	1,100 mg/kg	Note 97 Note 33
05.1.3	Cocoa-based spreads, including fillings	880 mg/kg	Note XS86 Note 33
14.1.5	Coffee, coffee substitutes, tea, herbal infusions, and other hot cereal and grain beverages, excluding cocoa	300 mg/kg	Note 160 Note 33
13.2	Complementary foods for infants and young children	4,400 mg/kg	Note 33 Note XS73 Note 230
14.1.2.3	Concentrates for fruit juice	1,000 mg/kg	Note 127 Note 33 Note 122 Note 40
14.1.3.3	Concentrates for fruit nectar	1,000 mg/kg	Note 127 Note 33 Note 122 Note 40
14.1.3.4	Concentrates for vegetable nectar	1,000 mg/kg	Note 127 Note 33 Note 40
01.3.1	Condensed milk (plain)	880 mg/kg	Note 33 Note 565
05.2	Confectionery including hard and soft candy, nougats, etc. other than food categories 05.1, 05.3 and 05.4	2,200 mg/kg	Note 33 Note XS309R
09.2.4.1	Cooked fish and fish products	2,200 mg/kg	Note 33
09.2.4.2	Cooked mollusks, crustaceans, and echinoderms	2,200 mg/kg	Note 33
04.2.2.8	Cooked or fried vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), and seaweeds	2,200 mg/kg	Note 613 Note 76 Note 33
07.1.2	Crackers, excluding sweet crackers	9,300 mg/kg	Note 229 Note 33
01.4	Cream (plain) and the like	2,200 mg/kg	Note 33 Note 617
01.7	Dairy-based desserts (e.g. pudding, fruit or flavoured yoghurt)	1,500 mg/kg	Note 610 Note 33
05.4	Decorations (e.g. for fine bakery wares), toppings (non-fruit) and sweet sauces	1,500 mg/kg	Note 33
13.5	Dietetic foods (e.g. supplementary foods for dietary use) excluding products of food categories 13.1 - 13.4 and 13.6	2,200 mg/kg	Note 33
13.3	Dietetic foods intended for special medical purposes (excluding products of food category 13.1)	2,200 mg/kg	Note 33 Note 566
13.4	Dietetic formulae for slimming purposes and weight reduction	2,200 mg/kg	Note 33
14.2.6	Distilled spirituous beverages containing more than 15% alcohol	440 mg/kg	Note 88 Note 33
06.4.2	Dried pastas and noodles and like products	900 mg/kg	Note 33 Note 211

Number	Food Category	Max Level	Notes
 04.2.2.2	Dried vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), seaweeds, and nuts and seeds	5,000 mg/kg	 Note 613  Note 76  Note 33
 01.8.2	Dried whey and whey products, excluding whey cheeses	4,400 mg/kg	 Note XS331  Note 33
 08.4	Edible casings (e.g. sausage casings)	1,100 mg/kg	 Note 33
 03.0	Edible ices, including sherbet and sorbet	7,500 mg/kg	 Note 33
 10.4	Egg-based desserts (e.g. custard)	1,400 mg/kg	 Note 33
 02.3	Fat emulsions mainly of type oil-in-water, including mixed and/or flavoured products based on fat emulsions	2,200 mg/kg	 Note 33
 02.2.2	Fat spreads, dairy fat spreads and blended spreads	2,200 mg/kg	 Note 33  Note 555  Note 530  Note 554
 02.4	Fat-based desserts excluding dairy-based dessert products of food category 01.7	1,500 mg/kg	 Note 33
 01.2	Fermented and renneted milk products (plain)	1,000 mg/kg	 Note 610  Note 33  Note 635
 04.1.2.10	Fermented fruit products	2,200 mg/kg	 Note 33  Note 636
 04.2.2.7	Fermented vegetable (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera) and seaweed products, excluding fermented soybean products of food categories 06.8.6, 06.8.7, 12.9.1, 12.9.2.1 and 12.9.2.3	2,200 mg/kg	 Note 572  Note 33  Note 636
 07.2	Fine bakery wares (sweet, salty, savoury) and mixes	9,300 mg/kg	 Note 229  Note 33
 09.3.1	Fish and fish products, including mollusks, crustaceans, and echinoderms, marinated and/or in jelly	2,200 mg/kg	 Note 33
 09.3.2	Fish and fish products, including mollusks, crustaceans, and echinoderms, pickled and/or in brine	2,200 mg/kg	 Note 33
 01.1.4	Flavoured fluid milk drinks	1,500 mg/kg	 Note 610  Note 33  Note 364
 06.2.1	Flours	2,500 mg/kg	 Note 640  Note 469  Note 33  Note 225
 01.1.3.2	Fluid buttermilk (plain)	1,500 mg/kg	 Note 227  Note 33  Note 397
 01.1.1	Fluid milk (plain)	1,500 mg/kg	 Note 227  Note 33
 13.6	Food supplements	2,200 mg/kg	 Note 33
 13.1.3	Formulae for special medical purposes for infants	450 mg/kg	 Note 551  Note 587  Note 33  Note 586  Note 230
 06.4.1	Fresh pastas and noodles and like products	2,500 mg/kg	 Note 33  Note 211
 09.2.2	Frozen battered fish, fish fillets, and fish products, including mollusks, crustaceans, and echinoderms	2,200 mg/kg	 Note 33  Note 299
 10.2.2	Frozen egg products	1,290 mg/kg	 Note 33
 09.2.1	Frozen fish, fish fillets, and fish products, including mollusks, crustaceans, and echinoderms	2,200 mg/kg	 Note XS292  Note XS191  Note XS312  Note 33  Note XS36 Note 394 Note 393

Number	Food Category	Max Level	Notes
 09.2.3	Frozen minced and creamed fish products, including mollusks, crustaceans, and echinoderms	2,200 mg/kg	 Note 33
 08.2.3	Frozen processed meat, poultry, and game products in whole pieces or cuts	2,200 mg/kg	 Note 33
 04.2.2.1	Frozen vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), seaweeds, and nuts and seeds	5,000 mg/kg	 Note 76  Note 33  Note 637
 04.1.2.11	Fruit fillings for pastries	1,500 mg/kg	 Note 33
 04.1.2.3	Fruit in vinegar, oil, or brine	2,200 mg/kg	 Note 33  Note 636
 14.1.2.1	Fruit juice	1,000 mg/kg	 Note 33  Note 122  Note 40
 14.1.3.1	Fruit nectar	1,000 mg/kg	 Note 33  Note 122  Note 40
 04.1.2.8	Fruit preparations, including pulp, purees, fruit toppings and coconut milk	350 mg/kg	 Note XS314R  Note 33
 04.1.2.9	Fruit-based desserts, including fruit-flavoured water-based desserts	1,500 mg/kg	 Note 33
 04.1.2.6	Fruit-based spreads (e.g. chutney) excluding products of food category 04.1.2.5	1,100 mg/kg	 Note 33  Note XS160
 09.4	Fully preserved, including canned or fermented fish and fish products, including mollusks, crustaceans, and echinoderms	2,200 mg/kg	 Note XS94  Note XS119  Note 33  Note 436  Note XS3
 08.2.2	Heat-treated processed meat, poultry, and game products in whole pieces or cuts	1,320 mg/kg	 Note 33  Note 289
 05.1.5	Imitation chocolate, chocolate substitute products	2,200 mg/kg	 Note 33
 13.1.1	Infant formulae	450 mg/kg	 Note 381  Note 551  Note 587  Note 33  Note 586  Note 230
 10.2.1	Liquid egg products	4,400 mg/kg	 Note 33  Note 67
 01.8.1	Liquid whey and whey products, excluding whey cheeses	880 mg/kg	 Note 228  Note 33
 14.2.5	Mead	440 mg/kg	 Note 88  Note 33
 01.5.2	Milk and cream powder analogues	4,400 mg/kg	 Note 482  Note 88  Note 33  Note 483
 01.5.1	Milk powder and cream powder (plain)	4,400 mg/kg	 Note 548  Note 547  Note 33  Note 544  Note 543
 07.1.6	Mixes for bread and ordinary bakery wares	9,300 mg/kg	 Note 229  Note 33
 08.2.1	Non-heat treated processed meat, poultry, and game products in whole pieces or cuts	2,200 mg/kg	 Note 33
 01.1.2	Other fluid milk (plain)	2,200 mg/kg	 Note 33  Note 364  Note 411
 07.1.3	Other ordinary bakery products (e.g. bagels, pita, English muffins)	9,300 mg/kg	 Note 229  Note 33

Number	Food Category	Max Level	Notes
 11.4	Other sugars and syrups (e.g. xylose, maple syrup, sugar toppings)	1,320 mg/kg	 Note 33
 04.2.1.3	Peeled, cut or shredded fresh vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), seaweeds, and nuts and seeds	5,600 mg/kg	 Note 76  Note 33
 11.1.2	Powdered sugar, powdered dextrose	6,600 mg/kg	 Note 33  Note 465  Note 56
 06.4.3	Pre-cooked pastas and noodles and like products	2,500 mg/kg	 Note 33  Note 475  Note 211
 10.3	Preserved eggs, including alkaline, salted, and canned eggs	1,000 mg/kg	 Note 33
 01.6.4	Processed cheese	9,000 mg/kg	 Note 33
 08.3	Processed comminuted meat, poultry, and game products	2,200 mg/kg	 Note XS88  Note 33  Note 302
 15.0	Ready-to-eat savouries	2,200 mg/kg	 Note 33
 09.3.3	Salmon substitutes, caviar, and other fish roe products	2,200 mg/kg	 Note 33  Note XS291
 12.1.1	Salt	8,800 mg/kg	 Note 33
 12.1.2	Salt Substitutes	4,400 mg/kg	 Note 33
 12.6	Sauces and like products	2,200 mg/kg	 Note 33  Note XS302  Note 573
 12.2.2	Seasonings and condiments	2,200 mg/kg	 Note 33  Note 226
 09.3.4	Semi-preserved fish and fish products, including mollusks, crustaceans, and echinoderms (e.g. fish paste), excluding products of food categories 09.3.1 - 09.3.3	2,200 mg/kg	 Note 33  Note 193
 09.2.5	Smoked, dried, fermented, and/or salted fish and fish products, including mollusks, crustaceans, and echinoderms	2,200 mg/kg	 Note XS236  Note XS167  Note 420  Note XS244  Note 33  Note XS311  Note XS189  Note 413  Note 334
 07.1.1.2	Soda breads	9,300 mg/kg	 Note 229  Note 33
 12.5	Soups and broths	1,500 mg/kg	 Note 343  Note 33
 06.8.3	Soybean curd (tofu)	100 mg/kg	 Note 33  Note 649
 06.8.1	Soybean-based beverages	1,300 mg/kg	 Note 647  Note 33
 12.9	Soybean-based seasonings and condiments	1,200 mg/kg	 Note 33  Note XS298R
 07.1.5	Steamed breads and buns	9,300 mg/kg	 Note 229  Note 33
 04.2.1.2	Surface-treated fresh vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), seaweeds, and nuts and seeds	1,760 mg/kg	 Note 16  Note 33
 11.6	Table-top sweeteners, including those containing high-intensity sweeteners	1,000 mg/kg	 Note 33
 01.6.1	Unripened cheese	4,400 mg/kg	 Note 487  Note 33  Note 496  Note 495  Note 562  Note 564

Number	Food Category	Max Level	Notes
 04.2.2.6	Vegetable (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), seaweed, and nut and seed pulps and preparations (e.g. vegetable desserts and sauces, candied vegetables) other than food category 04.2.2.5	2,200 mg/kg	 Note 33  Note XS57  Note XS259R  Note XS308R
 04.2.2.5	Vegetable (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), seaweed, and nut and seed purees and spreads (e.g., peanut butter)	2,200 mg/kg	 Note 76  Note 33  Note XS57
 04.2.2.3	Vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), and seaweeds in vinegar, oil, brine, or soybean sauce	2,200 mg/kg	 Note 638  Note 33  Note 636
 14.1.4	Water-based flavoured drinks, including "sport," "energy," or "electrolyte" drinks and particulated drinks	1,000 mg/kg	 Note 33

Note: Unless otherwise specified, food additive provisions apply to the food category indicated (e.g. Dairy), as well as to all subcategories of that category (e.g. Cheese, Ripened Cheese, etc.).

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